Basic Environment & Ecosystem Concepts

- **Boring's Elements of the Environment:** According to E.G. Boring, the environment consists of external conditions that influence an organism, which include three types of experiences:
    - *Physical Experiences:* Climate, light, sound, soil, water, air, and natural disasters.
    - *Biological Experiences:* Interactions with plants, animals, microorganisms, and food chains.
    - *Social Experiences:* Family structure, education, culture, economic conditions, and technology.
- **Douglas & Hallend's Components:** They defined the environment as the totality of external forces, influences, and conditions—both natural and man-made—that affect the life, growth, and development of organisms.
- **Body and Air Composition (Ancient Indian Knowledge):**
    - *Body Composition:* The human body is composed of the *Panchamahabhutas* (five elements): Earth (Prithvi) for solid structures like bones, Water (Apas) for fluids like blood, Fire (Agni) for metabolism, Air (Vayu) for breathing and movement, and Space (Akasha) for bodily cavities.
    - *Air Composition:* Ayurveda describes air philosophically as a balance of *Vata dosha*, which represents movement, dryness, and coldness.
- **Coniferous Forests:** These are narrow-leaved forests featuring flora such as *Pinus*, *Quercus*, and *Berberis*, and are home to fauna like the Himalayan goat, black bear, and sheep.
- **Global Warming:** It is the gradual overall increase in the Earth's average temperature caused by greenhouse gases (such as carbon dioxide, water vapor, nitrous oxide, and methane) trapping heat and light from the sun within the Earth's atmosphere.
- **Ecosystems in a Nutshell:** Yes, it is correct to state this. An ecosystem is defined as a biological community of interacting organisms (the living or biotic environment, such as plants and animals) and their physical surroundings (the non-living or abiotic environment, such as soil, air, and water) exchanging energy and matter.
- **Role of Photosynthesis:** No ecosystem is complete without photosynthesis because it is the fundamental process through which producers (autotrophs like plants) capture solar energy, carbon dioxide, and water to manufacture

food. This process converts radiant energy into chemical energy, which is then transferred through the food chain to sustain all consumers and decomposers.

- **Three Types of Biodiversity:**
    1. *Ecosystem Biodiversity:* The variety of different ecosystems (such as forests, wetlands, deserts, and oceans) and the ecological processes that occur within a specific region.
    2. *Species Diversity:* The number and variety of different plant and animal species present in a given area.
    3. *Genetic Diversity:* Though not explicitly defined as a separate heading in the text, it is frequently cited as the variation of genes within a species, which is crucial for ecological stability, resilience, and species adaptation.
- **Energy Flow in an Ecosystem:** Energy is primarily derived from the sun and flows in a unidirectional manner. It is captured by producers (plants) and then transferred to consumers (herbivores and carnivores) and ultimately to decomposers through complex food chains and food webs.

Biodiversity, Endemism & Conservation

- **Black Pepper & Spices Endemic Region:** They are endemic to the Western Ghats, India.
- **Native Land of Cardamom and Spices:** The Western Ghats region is recognized as the native land of cardamom and spices.
- **Native Land of Kahwa:** Kahwa is a traditional green tea grown in the valleys of Kashmir, India.
- **Indo-Burma Hotspot Countries:** This hotspot covers northeastern India (excluding Assam and the Andaman Islands), Myanmar, southern China, Cambodia, Vietnam, Laos, and Thailand.
- **Sundaland Biodiversity Hotspot:** This region includes the Malay Peninsula, Borneo, Sumatra, Java, the Nicobar Islands (India), and surrounding islands across Malaysia, Indonesia, Brunei, Singapore, and parts of the Philippines and southern Thailand.
- **Contribution of Endemic Species:** Endemic species enhance the unique biodiversity of hotspots like the Western Ghats, making these regions vital for global conservation efforts and maintaining region-specific ecosystem services.
- **National Forest Policy (1988) Target:** The policy targets bringing 33% of India's geographical area under forest cover (with a higher target of 66% or 67% in hilly regions).

- **In situ (Natural) Method of Biodiversity Conservation:** *In situ* conservation is the practice of protecting species within their natural habitats, such as in national parks, wildlife sanctuaries, and biosphere reserves. This method preserves entire ecosystems, maintains ecological balance by protecting species interactions, and safeguards genetic diversity. It is also cost-effective and culturally inclusive, often utilizing indigenous knowledge.
- **Wetlands as Abused Aquatic Habitats:** Aquatic habitats like wetlands are highly abused due to intense development pressures. For example, freshwater floodplain swamps and wetlands are frequently destroyed by being drained to make way for agricultural expansion, such as wet rice cultivation.

Water Resources, Filtration & Pollution

- **Why do aquatic animals die?** Aquatic animals primarily die due to **water pollution and oxygen depletion**. This includes the discharge of **water and thermal pollution from industrial effluents** into aquatic ecosystems.
- **What are the effects of thermal pollution on aquatic life?** Thermal pollution raises water temperatures, which **decreases the water's oxygen-carrying capacity and causes hypoxia (low oxygen)**. This stresses aquatic organisms, disrupts their breeding patterns, accelerates their metabolism, and can lead to the suffocation and death of temperature-sensitive species. It also disrupts the food chain, encourages invasive species, and can cause coral bleaching.
- **What is the cause of dropping down the water table (groundwater level)?** The water table drops due to several factors, including:
  - **Overextraction:** Excessive pumping for agricultural, industrial, and domestic use.
  - **Deforestation:** Removing trees reduces the infiltration of rainwater into the soil, as plant roots help hold water in the upper layers of the soil.
  - **Urbanization:** Impervious surfaces increase surface runoff rather than allowing water to absorb into the ground.
  - **Climate Change and Inefficient Irrigation:** Reduced rainfall, increased evaporation, and excessive water use in farming also lower the water table.
- **Which technological invention is extensively used for water management in arid regions of Asia?** Both **drip irrigation** and **desalination technology** are extensively used. Drip irrigation is widely

applied to conserve water and increase crop productivity in arid areas like Israel and parts of India. Desalination technology is also highlighted as a primary solution for managing water in arid Asian regions.

- **Mention water conflict issues within Asian countries.** Water conflicts in Asia stem from growing demand, transboundary river systems, and uneven distribution. Major issues include:
    - **Inter-State Disputes in India:** Conflicts between states, such as the Cauvery Water Dispute (Tamil Nadu vs. Karnataka) and disputes over the Krishna and Ravi-Beas rivers.
    - **India–China Conflict:** Tensions over the Brahmaputra River due to Chinese upstream hydropower projects, water diversion plans, and a lack of transparent hydrological data.
    - **India–Pakistan:** Friction surrounding the Indus Water Treaty, exacerbated by Indian infrastructure projects like the Kishanganga and Ratle dams in Kashmir.
    - **China–Southeast Asia:** Disputes over the Mekong River, where China's upstream dams disrupt seasonal flows, severely affecting agriculture and fisheries in downstream nations like Laos, Thailand, Cambodia, and Vietnam.
    - **Central Asia:** Disagreements among former Soviet republics sharing the Aral Sea Basin over whether to prioritize water for irrigation or hydropower.
- **How can we trap smaller contaminants in water?** Smaller contaminants can be trapped using **ultrafiltration**, which removes viruses, proteins, and organic molecules using moderate pressure, and **nanofiltration**, which can remove bivalent ions.
- **If Reverse Osmosis (RO) can get everything out that would make water undrinkable, justify why RO by itself might not be the best solution.** RO might not be the best standalone solution because it comes with a **high cost due to the high pressure required**. Furthermore, large particles can plug the RO membrane, and RO can be an "overkill" solution for water that doesn't have salt that needs to be removed (such as fresh or lake water).
- **How do you determine what filtration method to use to remove contaminants in a water sample?** The appropriate filtration method is determined by a water quality test evaluating three main factors:

1. **Type and Size of Contaminants:** Whether the water contains particulate matter, microbial contaminants, chemical pollutants, or specific sizes of particles.
2. **Water Source:** Whether it is groundwater, surface water, or wastewater.
3. **Purpose of Filtration & Cost:** How the water will be used (e.g., drinking, industrial use, or irrigation) and the relative cost of the filtration method.

- **What are two benefits that nanomembranes bring to the filtration of water?**
    0. **Lower Cost:** Because their pores are close in size to RO filters, they offer a **cheaper alternative to RO** and can operate at a low operating cost.
    1. **Charge-Based Filtration:** Advances in nanotechnology allow nanomembranes to be built in layers that can **filter not only based on size but also based on electrostatic charge**, stopping specific small particles while allowing water to pass through.
- **What is eutrophication?** Eutrophication is **nutrient overloading in water bodies** which causes massive algal blooms and subsequent oxygen depletion in the water.

Agriculture, Soil & Land Management

- **What is vermicomposting?** *(Note: This information is not explicitly defined in your provided sources, though the sources do mention it as a type of biofertilizer).* Vermicomposting is the process of using various species of worms, usually red wigglers or earthworms, to break down organic waste material into a nutrient-rich, dark, and soil-like substance known as vermicompost or worm castings, which is used as a highly effective natural fertilizer.
- **State one major reason for soil degradation in agricultural lands.** One major reason is the **excessive and continuous use of chemical fertilizers and pesticides**, which harms beneficial microorganisms, degrades soil fertility, and damages the physical structure of the soil.
- **What is the role of biofertilizers in the prevention of soil degradation?** Biofertilizers **provide necessary nutrition to plants without reducing the fertility of the soil**, unlike chemical fertilizers which degrade soil health over time.

- **How do human activities trigger landslides?** Human activities such as **deforestation, mining, quarrying, road construction, and unplanned urbanization** destabilize hilly terrains. Removing vegetation eliminates the root systems that hold soil together, while construction overloads slopes and improper drainage systems cause water to accumulate, severely reducing soil cohesion and triggering landslides during heavy rainfall.
- **What is wasteland reclamation, and what are two methods of reclaiming wasteland?** Wasteland reclamation is the **process of restoring degraded or unproductive land** so that it can be used for ecological or agricultural purposes. Two effective methods are:
    1. **Afforestation:** Planting trees to restore ecological balance, improve soil structure, and prevent erosion.
    2. **Soil Treatment / Amendment:** Adding organic matter or chemicals (such as adding gypsum or lime to saline or alkaline soils) to correct nutrient deficiencies and improve soil texture. *(Contour bunding is also mentioned as a method to reduce erosion and runoff)*
- **In which country has precision agriculture technology rapidly advanced to optimize natural resource use? Japan** is specifically noted as a country where precision agriculture technology has rapidly advanced to optimize natural resource use. (China and India are also heavily implementing these technologies).

Technology, Energy & Mineral Resources

- **How Information Technology (IT) helps in improving human health:** IT improves human health primarily through **telemedicine and health monitoring apps**.
- **Two ways IT has improved public healthcare services:** IT has significantly improved public healthcare services through the implementation of **telemedicine** and **electronic health records (EHRs)**.
- **How IT helps in controlling pollution:** IT enables **real-time data collection and monitoring systems** that track air and water pollution levels to support environmental decision-making.
- **Role of mobile health apps:** Mobile health apps aid in disease prevention and awareness by providing **health tips, tracking symptoms, and spreading awareness** about healthy habits and disease prevention.

- **Use of remote sensing in environmental monitoring:** Remote sensing is used to **detect deforestation, monitor land use changes, track urban sprawl, and assess natural disasters** by collecting aerial or satellite data.
- **Technological innovation for Asian forest monitoring: Remote sensing, Geographic Information Systems (GIS), drones, and satellite imagery** are extensively used in Asia for monitoring and managing forest resources.
- **Largest producer of rare earth elements: China** is the largest producer of rare earth elements, which are crucial for technological innovations.
- **Technologies for mineral exploration in Asia: Remote sensing and GIS technologies** are prominently used for mineral exploration across Asia.
- **Emerging solar photovoltaic panel manufacturer: China** leads globally in solar capacity and panel deployments, acting as a major hub for solar power expansion and integration.
- **Most abundantly harnessed renewable energy in Southeast Asia: Hydropower** is the most abundantly harnessed renewable energy source in the Southeast Asian region.
- **Resources found in Sri Lanka and the Malabar Coast:** The **Malabar Coast** of India is endemic to **black pepper**. **Sri Lanka** is known for its rich **tea** cultivation in the uplands and is a major producer of **coconuts**.
- **Effect of plastic burning:** Burning plastic releases **toxic gases that cause severe air pollution**.

Human Rights, Laws & International Organizations

- **Foundation of international human rights law:** The **Universal Declaration of Human Rights (UDHR)** is considered the foundation of international human rights law.
- **How the UDHR ensures the right to life/liberty:** Under **Article 3**, the UDHR explicitly states that "everyone has the right to life, liberty, and security of person".
- **Organization monitoring global human rights violations:** The **UNHRC (United Nations Human Rights Council)** primarily monitors human rights violations at a global level.
- **Organization focusing on child welfare: UNICEF** focuses on child welfare and children's rights at the international level.

- **Two major rights guaranteed to children (UN Convention):** The UN Convention on the Rights of the Child guarantees the **right to education** and the **right to protection from abuse and exploitation**.
- **International body overseeing global environmental protection:** The **IUCN (International Union for Conservation of Nature)** is a key global organization working to conserve nature, protect species, and ensure the sustainable use of resources.
- **Role of the National Human Rights Commission (NHRC) in India:** The NHRC **investigates human rights violations, advises the government** on related matters, and **promotes awareness and education** on human rights across the country.